Last time: If $\Delta u = 0$, then $u \in C^\infty$.

Even better, if $u \in C(U)$ and satisfies MVP, then $u \in C^\infty$ and $\Delta u = 0$.

Today: <u>quantitative</u> regularity

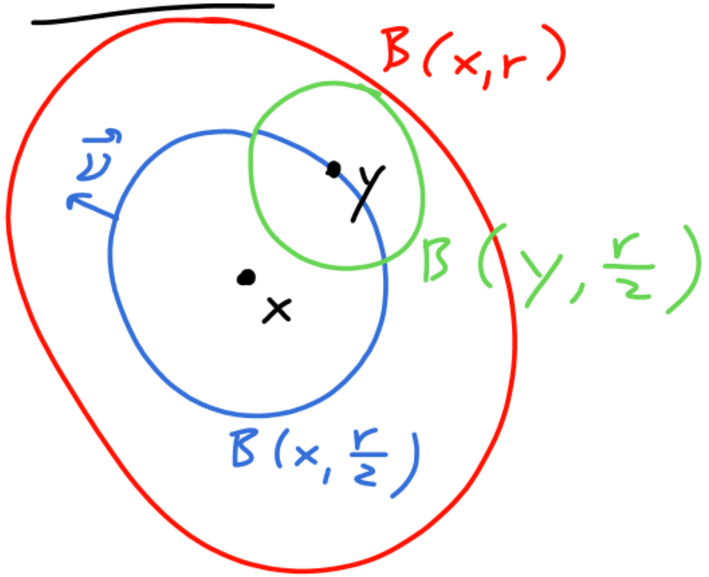
<u>Thm</u> If u is harmonic in $U \subset \mathbb{R}^n$ and $\overline{B(x,r)} \subset U$, and $D^\beta u$ is a partial derivative of u of order k, then

$$|D^\beta u(x)| \leq \frac{C(n,k)}{r^{n+k}} \int_{B(x,r)} |u|\,dy,$$

$$C(n,k) = \begin{cases} \dfrac{(2^{n+1} n k)^k}{\alpha(n)}, & k \geq 1 \\ \dfrac{1}{\alpha(n)}, & k = 0 \end{cases}$$

$\alpha(n)$ = Volume of $B(0,1)$ in dimension n.

Proof $k=1$ case:



$$\Delta u = 0$$

$$\frac{\partial}{\partial x_i}(\Delta u) = 0$$

$$\Delta(u_{x_i}) = 0 \longrightarrow u_{x_i} \text{ is harmonic}$$

$$u_{x_i}(x) = \fint_{B(x,r/2)} u_{x_i}(y)\,dy = \frac{1}{\alpha(n)(r/2)^n}\int_{B(x,r/2)} u_{x_i}(y)\,dy$$

$$= \frac{1}{\alpha(n)\left(\frac{r}{2}\right)^n}\int_{B(x,r/2)} \operatorname{div}(F)\,dy$$

$$F = u\cdot\begin{bmatrix}0\\0\\\vdots\\1\\0\\0\\\vdots\\0\end{bmatrix} \leftarrow \text{ith position}$$

$$\operatorname{div} F = u_{x_i}$$

$$= \frac{1}{\alpha(n)\left(\frac{r}{2}\right)^n}\int_{\partial B(x,r/2)} F\cdot\vec{\nu}\,dS$$

$$= \frac{1}{\alpha(n)\left(\frac{r}{2}\right)^n}\int_{\partial B(x,r/2)} u\,\nu_i\,dS$$

$$|u_{x_i}(x)| = \frac{1}{\alpha(n)\left(\frac{r}{2}\right)^n}\left|\int_{\partial B(x,r/2)} u\nu_i \, dS\right|$$

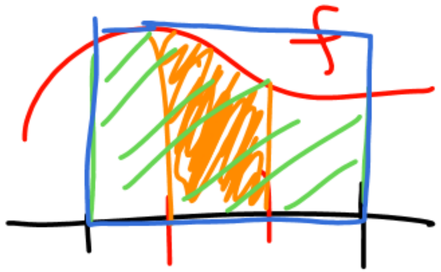
$$\boxed{\left|\int f\,dx\right| \le \int |f|\,dx}$$

$$\le \frac{1}{\alpha(n)\left(\frac{r}{2}\right)^n}\int_{\partial B(x,r/2)} |u|\,dS \quad (*)$$

For any $y \in \partial B(x, r/2)$, $\boxed{B(y, r/2) \subseteq B(x,r)}$

Mean Value Property again:

$$|u(y)| = \left|\fint_{B(y,\frac{r}{2})} u(z)\,dz\right| = \frac{1}{\alpha(n)\left(\frac{r}{2}\right)^n}\left|\int_{B(y,r/2)} u(z)\,dz\right|$$



$$\le \frac{1}{\alpha(n)\left(\frac{r}{2}\right)^n}\int_{B(y,r/2)} |u|\,dz \le \frac{1}{\alpha(n)\left(\frac{r}{2}\right)^n}\int_{B(x,r)} |u|\,dz$$

Use this in $(*)$

$$|u_{x_i}(x)| \le \frac{1}{\alpha(n)\left(\frac{r}{2}\right)^n} \int_{\partial B(x,\frac{r}{2})} |u| \, dS \le \frac{1}{\cancel{\alpha(n)}\left(\frac{r}{2}\right)^n} \boxed{\max_{\partial B(x,\frac{r}{2})} |u|} \, \cancel{\alpha(n)} n \left(\frac{r}{2}\right)^{n-1}$$

$$\le \frac{n}{r/2} \frac{1}{\alpha(n)\left(\frac{r}{2}\right)^n} \int_{B(x,r)} |u| \, dz$$

$$\le \frac{2^{n+1} n}{\alpha(n) r^{n+1}} \int_{B(x,r)} |u| \, dz$$

Sketch of higher derivative estimates:

Use Proof By Induction

Base case: $k=0$ or $k=1$ ✓

Inductive step: Assume thm is true for some $k \geq 1$,

$$|D^\beta u(x)| \leq \frac{C(n,k)}{r^{n+k}} \int_{B(x,r)} |u|\, dz \quad \text{if } |\beta| \leq k$$

use this to show $|D^\gamma u(x)| \leq \dfrac{C(n,k+1)}{r^{n+k+1}} \displaystyle\int_{B(x,r)} |u|\, dz, \quad |\gamma| = k+1.$

(See Evans for details)

Two consequences:

<u>Thm</u> (Liouville's Thm)

If u is harmonic in all of $\mathbb{R}^n$, and bounded, then u is constant.

<u>Pf</u> If u is harmonic in $\mathbb{R}^n$, and $|u| \leq M$,

By Previous Thm ($k=1$ case).

$$|Du(x)| \leq \frac{C(n,1)}{r^{n+1}} \int_{B(x,r)} |u| \, dy \leq \frac{C(n,1)}{r^{n+1}} \int_{B(x,r)} M \, dy = \frac{C(n,1)\alpha(n) M r^n}{r^{n+1}}$$

$$= \frac{C(n,1)\alpha(n) M}{r}$$

$$|Du(x)| \le \frac{C(n,1)\,\alpha(n)\,M}{r} \quad \text{For any } r > 0.$$

Send $r \to \infty$: $|Du(x)| = 0$

$\Rightarrow$ $Du(x) = 0$ for all $x \in \mathbb{R}^n$ $\Rightarrow$ u is constant.

<u>Thm</u> Harmonic Functions are (real) analytic,

i.e. the Taylor series

$$\sum_{k=0}^{\infty} \sum_{|\beta|=k} \frac{D^\beta u(x_0)\,(x-x_0)_1^{\beta_1} \cdots (x-x_0)_n^{\beta_n}}{k!} \longrightarrow u(x)$$

for all x.

<u>Pf</u> See Theorem 10, Ch 2, Evans.

Harnack Inequality

Idea: The ratio of $\frac{u(x)}{u(y)}$ is controlled from above & below for any $x, y \in U$.

"u is not oscillating too much"

this

not this

<u>Thm</u> (Harnack Inequality)

Let $u > 0$ be harmonic in $U \subseteq \mathbb{R}^n$, and let V be an open set with $\overline{V} \subset U$. Then, there is a constant C depending only on V and d, so that

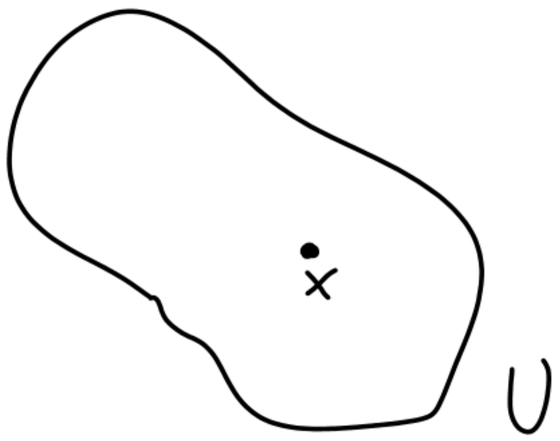
$$\frac{1}{C} u(x) \leq \underline{u(y) \leq C u(x)} \quad \text{for all } x, y \in V.$$

(Equivalent to $u(y) \leq C u(x) \ \forall x, y$

or $\dfrac{u(y)}{u(x)} \leq C \ \forall x, y$)

Interpretation:

Consider $\begin{cases} \Delta u = 0 & \text{in } U \\ u = g & \text{on } \partial U \end{cases}$



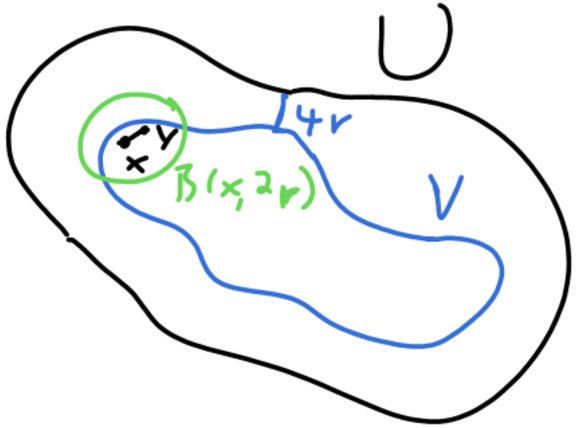
If $g \geq 0$, then $u \geq 0$.

If $u(x) = 0$ at any interior point, then $u \equiv 0$ in U by Strong Max Principle

Harnack is a quantitative version of this:

If $u(x)$ is close to 0 somewhere, it is close to 0 everywhere in U.

Proof

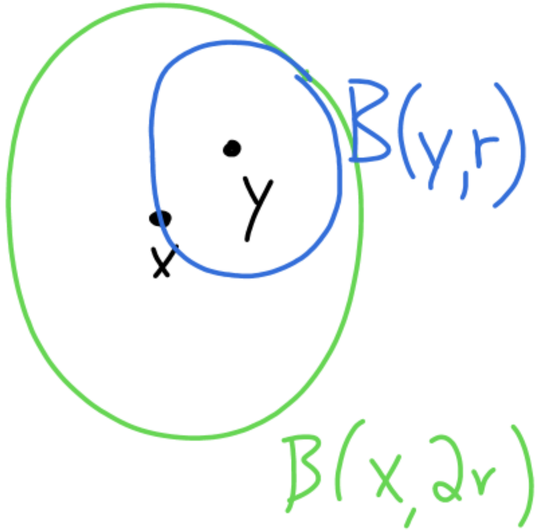


Let $r = \frac{1}{4}\,\mathrm{dist}(V, \partial U)$

Take $x, y \in V$ with $|x-y| = r$.

Mean Value Property:

$$u(x) = \fint_{B(x,2r)} u(z)\,dz = \frac{1}{\alpha(n)(2r)^n}\int_{B(x,2r)} u(z)\,dz \geq \frac{1}{\alpha(n)(2r)^n}\int_{B(y,r)} u(z)\,dz$$



$$= \frac{1}{2^n}\,\frac{1}{\alpha(n)r^n}\int_{B(y,r)} u(z)\,dz$$

$$= \frac{1}{2^n}\fint_{B(y,r)} u\,dz = \frac{1}{2^n}u(y).$$

We've shown:

If $|x-y| = r = \frac{1}{4}\,\mathrm{dist}(V, \partial U)$,

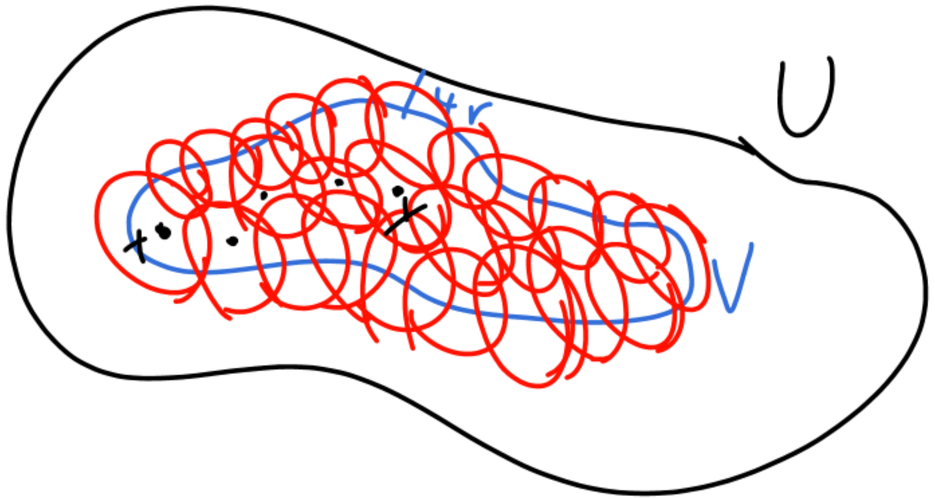
then $u(x) \geq \frac{1}{2^n} u(y)$ or $\boxed{u(y) \leq 2^n u(x).}$

Fact: There is $K > 0$, depending on V,

so that V can be covered by

K balls of radius r, with each ball

a subset of U.



For any $x, y \in V$, there are

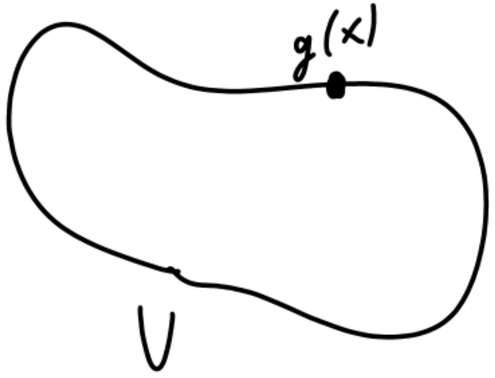
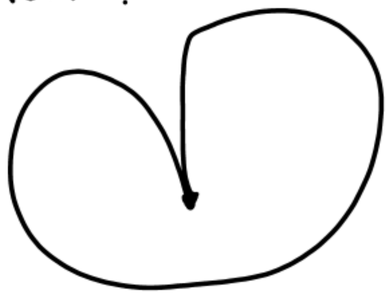
$x_1, \ldots, x_m \in V$, with $x_1 = x$, $x_m = y$,

each x_j, x_{j+1} contained in one of the balls.

Apply previous estimate:

$$u(x) \le 2^n u(x_1) \le 2^{2n} u(x_2) \le \cdots \le \underbrace{\boxed{2^{kn}}}_{= C} u(y)$$

We've been studying <u>interior</u> regularity for harmonic functions



Regularity up to ∂U may not hold, if the boundary is irregular.